TIPPECC Summer School
on the Analysis of Climate Model Data

Introduction to the Climate Data Operators (CDO)

Dr. Torsten Weber & Franziska Zander

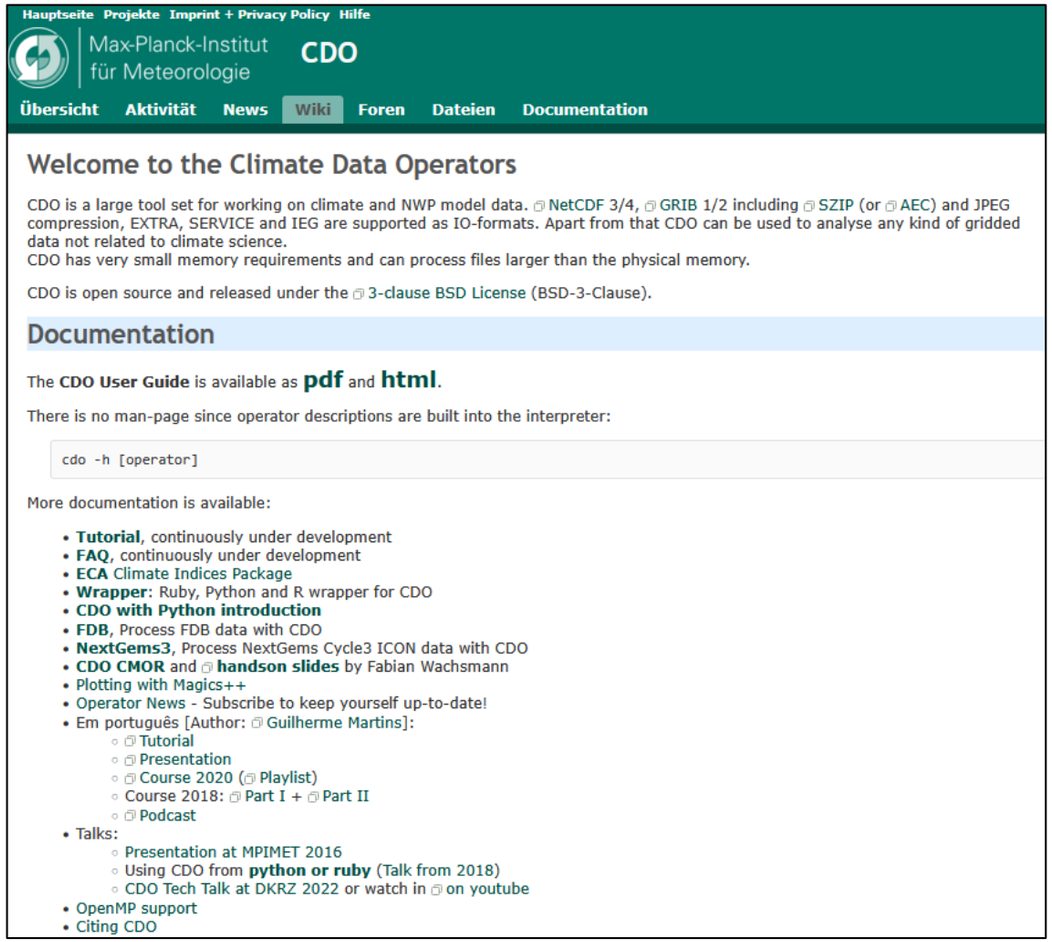
WITS Rural Facility

09 – 12 March 2026

With funding from the:
Federal Ministry of Research, Technology and Space

Aim of the lesson

- To make you familiar with the CDO
- Introduction of important CDO commands
- First steps how to analyse climate model data
  - time series data
  - spatial data
  - extract data into an ASCII file



Hauptseite Projekte Imprint + Privacy Policy Hilfe

Max-Planck-Institut für Meteorologie CDO

Übersicht Aktivität News Wiki Foren Dateien Documentation

Welcome to the Climate Data Operators

CDO is a large tool set for working on climate and NWP model data. NetCDF 3/4, GRIB 1/2 including SZIP (or AEC) and JPEG compression, EXTRA, SERVICE and IEG are supported as IO-formats. Apart from that CDO can be used to analyse any kind of gridded data not related to climate science.
CDO has very small memory requirements and can process files larger than the physical memory.

CDO is open source and released under the 3-clause BSD License (BSD-3-Clause).

Documentation

The **CDO User Guide** is available as **pdf** and **html**.

There is no man-page since operator descriptions are built into the interpreter:

```
cdo -h [operator]
```

More documentation is available:

- **Tutorial**, continuously under development
- **FAQ**, continuously under development
- **ECA** Climate Indices Package
- **Wrapper**: Ruby, Python and R wrapper for CDO
- **CDO with Python introduction**
- **FDB**, Process FDB data with CDO
- **NextGems3**, Process NextGems Cycle3 ICON data with CDO
- **CDO CMOR** and **handson slides** by Fabian Wachsmann
- Plotting with Magics++
- Operator News - Subscribe to keep yourself up-to-date!
- Em português [Author: Guilherme Martins]:
  - Tutorial
  - Presentation
  - Course 2020 (Playlist)
  - Course 2018: Part I + Part II
  - Podcast
- Talks:
  - Presentation at MPIMET 2016
  - Using CDO from **python or ruby** (Talk from 2018)
  - CDO Tech Talk at DKRZ 2022 or watch in on youtube
- OpenMP support
- Citing CDO

(source: https://code.mpimet.mpg.de/projects/cdo/wiki/Cdo#Documentation)

Climate Data Operators (CDO)

What are the Climate Data Operators?

- command-line software tool used to analyze, manipulate, and process (large) climate and meteorological data stored in formats like NetCDF and GRIB.
- developed by MPI in Hamburg
- provides hundreds of operators to process gridded data

- CDO is freely available for Linux (Windows via WSL) (apt get install cdo OR (better) use a docker container or VM to avoid lib conflicts)

- A detailed documentation with a tutorial and the software can be find at https://code.mpimet.mpg.de/projects/cdo/
-

- For this course, a CDO manual and a CDO reference-card is stored in *~/training/code/cdo*

Climate Data Operators (CDO)

- **What can CDO do?**
  - Temporal operations
  - Spatial Operations
  - Statistics
  - Format conversions
  - Derived calculations

- **Why Is CDO So Popular?**
  - Extremely fast (written in C)
  - Works directly on large files (good for HPC)
  - Scriptable in Bash
  - CF-convention aware
  - Designed specifically for climate data

CDO Reference Card - excerpt

Syntax

cdo [Options] **Operator1** [**–Operator2** [**–OperatorN**]]

Options

-a	Generate an absolute time axis
-b <*nbits*>	Set the number of bits for the output precision (I8/I16/I32/F32/F64 for nc1,nc2,nc4,nc4c; F32/F64 for grb2,srv,ext,ieg; 1-24 for grb1,grb2) Add L or B for Little or Big endian byteorder
-f <*format*>	Outputformat: grb1,grb2,nc1,nc2,nc4,nc4c,srv,ext
-g <*grid*>	Grid or file name Grid names: `r<NX>x<NY>`, `n<N>`, `gme<NI>`
-h	Help information for the operators
-M	Indicate that the I/O streams have missing values
-m <*missval*>	Set the default missing value (default: `-9e+33`)
-O	Overwrite existing output file, if checked
-R	Convert GRIB1 data from reduced to regular grid
-r	Generate a relative time axis
-s	Silent mode
-t <*table*>	Set the parameter table name or file Predefined tables: echam4 echam5 mpiom1
-V	Print the version number
-v	Print extra details for some operators
-z szip	SZIP compression of GRIB1 records

Examples

```
cdo -V

cdo -h

cdo -a copy input_file output_file

cdo -f grb copy ifile.nc ofile.grb
```

CDO Reference Card

Operators

Information

info	Dataset information listed by parameter identifier
infon	Dataset information listed by parameter name
cinfo	Compact information listed by parameter name
map	Dataset information and simple map
<*operator*> `infiles`	

sinfo	Short information listed by parameter identifier
sinfon	Short information listed by parameter name
<*operator*> `infiles`	

xsinfo	Extra short information listed by parameter name
xsinfop	Extra short information listed by parameter identifi
<*operator*> `infiles`	

diff	Compare two datasets listed by parameter id
diffn	Compare two datasets listed by parameter name
<*operator*>[,*parameter*] `infile1 infile2`	

npar	Number of parameters
nlevel	Number of levels
nyear	Number of years
nmon	Number of months
ndate	Number of dates
ntime	Number of timesteps
ngridpoints	Number of gridpoints
ngrids	Number of horizontal grids
<*operator*> `infile`	

File operations

apply	Apply operators on each input file.
apply,*operators* `infiles`	

copy	Copy datasets
clone	Clone datasets
cat	Concatenate datasets
<*operator*> `infiles outfile`	

tee	Duplicate a data stream
tee,*outfile2* `infile outfile1`	

pack	Pack data
pack[,*parameter*] `infile outfile`	

replace	Replace variables
replace `infile1 infile2 outfile`	

duplicate	Duplicates a dataset
duplicate[,*ndup*] `infile outfile`	

mergegrid	Merge grid
mergegrid `infile1 infile2 outfile`	

merge	Merge datasets with different fields
merge `infiles outfile`	
mergetime	Merge datasets sorted by date and time
mergetime[,*options*] `infiles outfile`	

Climate Data Operators (CDO)

Examples

input file – green
output file – blue

- To change relative time to absolute time and select the year 1950 from a file:
  ```
  cdo -a selyear,1950 input_file output_file
  ```

- To select the single years 1950, 1960, 1970 from a file:
  ```
  cdo selyear,1950,1960,1970 input_file output_file
  ```

- To select the period from 1950 - 1970 from a file:
  ```
  cdo selyear,1950/1970 input_file output_file
  ```

- To select a geographical region from a file:
  Operator *sellonlatbox,lon_min,lon_max,lat_min,lat_max*

  South African region: lon. 8 deg. to 42 deg., lat. -38 deg. to -4 deg.
  ```
  cdo sellonlatbox,8,42,-38,-4 input_file output_file
  ```

Climate Data Operators (CDO)

Examples

input file – green
output file – blue

- To change the unit of monthly mean precipitation from [kg m-2 s-1] to [mm day-1], the following assumption must be made: 1 [kg m-2 s-1] ≈ 1 [litre m-2 s-1] ≈ 1 [mm s-1]:

```
cdo mulc,86400 pr_mon_file tmp_file (multiplied with seconds per day)
cdo chunit,"kg m-2 s-1","mm day-1" tmp_file result_file
```

- To change the unit of monthly mean precipitation from [mm day-1] to [mm month-1], use this:

```
cdo muldpm pr_mon_file tmp_file (multiplied days per months)
cdo chunit,"mm day-1","mm month-1" tmp_file result_file
```

- To calculate yearly sum from monthly precipitation [mm month-1], use this:

```
cdo yearsum pr_mon_file tmp_file
cdo chunit,"mm month-1","mm year-1" tmp_file result_file
```

Climate Data Operators (CDO)

Use pipes to combine several operations in one command

Syntax: *cdo op3 -op2 -op1 input_file output_file*

Example

To change the relative time to absolute time, select the period from 1950 - 1970 and select a region from a file:

```
cdo -a sellonlatbox,8,42,-38,-4 -selyear,1950/1970 input_file output_file
       second operation            first operation
```

Climate Data Operators (CDO)

Examples

input file – green
output file – blue

- To calculate a map of a climate change signal, the time mean of the historical data has to be subtracted from the time mean of the scenario data:

```
cdo timmean scen_2071-2100_file scen_tmp_file
cdo timmean hist_1971-2000_file hist_tmp_file
cdo sub scen_tmp_file hist_tmp_file result_file
```

  or same result using the *pipe* function:

```
cdo sub -timmean scen_2071-2100_file -timmean hist_1971-2000_file result_file
```

- Another example using the *pipe* function:
  Calculating the yearly sum from monthly mean precipitation [kg m-2 s-1] to [mm year-1]

```
cdo chunit,"kg m-2 s-1","mm year-1" -yearsum -muldpm -mulc,86400
                                                    pr_mon_file results_file
```

Climate Data Operators (CDO)

Examples

input file – green
output file – blue

- To calculate a yearly time series, the data has to be converted to year mean and field mean:
```
cdo fldmean -yearmean scen_2010-2100_file result_file
```

- To calculate a climate change signal for a time series, the time mean and field mean of historical data (-> single value) have to be subtracted from the field mean of the scenario data:
```
cdo fldmean -timmean hist_1971-2000_file mean_1971-2000_file
cdo sub -fldmean scen_2071-2100_file mean_1971-2000_file result_file
```

- To calculate a yearly time series as 30-years running mean, the data has to be converted to year mean and field mean as well as to running mean:
```
cdo runmean,30 -fldmean -yearmean scen_2010-2100 result_file
```

Climate Data Operators (CDO)

input file – green
output file – blue

Data masking

To order to cut specific areas such as basins or countries from the data, there is a CDO command called *ifthen*. This command needs a mask with zeros/ones at the same grid as the data.

Usage:

```
cdo ifthen mask_file input_file out_file
```

The mask_file must have the same grid and data format as the input file!

Example:

```
cdo ifthen land_sea_mask.nc precip.nc south_africa_precip.nc
```

Climate Data Operators (CDO)

How to extract point data from model data in an ASCII file?

input file – green
output file – blue

1. Calculate the weighted average of each grid point plus the 8 surrounding points to avoid outliers:
```
cdo smooth9 input_file tmp1_file
```

2. Perform the nearest neighbour remapping of the field to extract a longitude/latitude point:
```
cdo remapnn,lon=XX/lat=YY tmp1_file tmp2_file
```

3. Print a customised table in an ASCII file:
```
cdo outputtab,name,year,month,day,lon,lat,value tmp2_file >result.txt
```

Example: Output for temperature with longitude 10 deg. and latitude 20 deg.:

```
#     name   year month day     lon     lat     value
      tas    1980  1     16      10      20      281.863
      tas    1980  2     16      10      20      285.643
      tas    1980  3     16      10      20      292.52
      tas    1980  4     16      10      20      298.133
      .....
```

Climate Data Operators (CDO)

Practice

1. Change into the directory /home/user1/training/code/cdo/practice
2. Cut out the African continent from the global file “t2m_ERA5_global.2021_2025.nc” (“cdo sellonlatbox,…”).
3. Display the variable “2 m temperature (2tm)” over the African continent (“ncview …).
4. Use the corresponding button to navigate through the timeframes.
5. Calculate the annual mean “2tm” for Africa for the period 2021 to 2025 (“cdo yearmean input.nc output.nc” …).
6. Display the mean “2tm” over the African continent for the period 2021 to 2025 (“ncview …).
7. Calculate the annual (field mean) “2tm” over the Africa continent for the period 2021 to 2025 (cdo fldmean –yearmean …).
8. Display the field mean “2tm” over the African continent for the period 2021 to 2025 (“ncview …).

Notice: Change the directory using the command (“cd …”).
Coordinates of the African continent: longitude -20 to 53, latitude -36 to 40.

When to use CDO vs xarray (Python)?

Task Type:	Tool
Simple climate operator	CDO
Very Large files	CDO
HPC processing	CDO
Custom analysis	Python
Statistics	Python
Plotting	Python
ML	Python

CDO with Python

Thank you for your attention!